

# **Lecture 02 — Vector arithmetic and geometry**

Wed Aug 26

Last updated: August 20, 2026.

Reading: Boyd 1.2–1.3. Related objectives: [Lectures](#). [Schedule](#).

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

## Learning objectives

By the end of class, students should be able to:

1. Add, subtract, and scale vectors.
2. Interpret those operations geometrically.
3. Plot vectors in  $\mathbb{R}^2$  and recognize corresponding operations in  $\mathbb{R}^3$ .
4. Check whether an answer has sensible dimensions and units.

## Warm-up

Yesterday's highs (Mon–Thu), in °F, were

$$t = (63, 59, 61, 66).$$

What is  $n$ ? What is  $t_2$ , and what does it mean? Is 61 a scalar or a vector? Write the 4-vector of “each day five degrees cooler than  $t$ ” if you already know how — otherwise leave it blank and we will return to it.

## Fill in

If  $a$  and  $b$  are  $n$ -vectors, the sum  $a + b$  is the  $n$ -vector whose  $i$ th entry is \_\_\_\_\_. Addition is only defined when \_\_\_\_\_.

The identity for addition is \_\_\_\_\_:  $a + 0 = a$ . The difference  $a - b$  is \_\_\_\_\_.

If  $\beta$  is a scalar, the scalar–vector product  $\beta a$  is the  $n$ -vector whose  $i$ th entry is \_\_\_\_\_. In particular,  $(-1)a$  is written \_\_\_\_\_.

Geometry in  $\mathbb{R}^2$ : draw  $a$  as an arrow. Then  $\beta a$  with  $\beta > 0$  points \_\_\_\_\_ and has length scaled by \_\_\_\_\_. If  $\beta < 0$ , the arrow points \_\_\_\_\_. The sum  $a + b$  can be drawn tip-to-tail or as the diagonal of a \_\_\_\_\_.

The same formulas work entrywise in  $\mathbb{R}^3$ ; we just cannot sketch as easily on paper.

Adding a scalar to a vector is \_\_\_\_\_ in the mathematics of this course (even if some software will do it). Concatenating  $(1, 2)$  with  $(3, 4)$  is also \_\_\_\_\_ vector addition.

## Worked in class

**W1.** A point on the quad is  $p = (2, 1)$  (east, north, in 10 m units). A walk is the displacement  $a = (3, -1)$ . Compute  $p + a$  and sketch  $p$ ,  $a$ , and  $p + a$  in  $\mathbb{R}^2$ . Then compute  $a - p$ . Does  $a - p$  have the same units as  $p$ ?

**W2.** Two rain gauges, in inches, over three days:

$$r = (0.40, 0, 1.10), \quad s = (0.15, 0.20, 0.30).$$

Compute  $r + s$  and  $2r$ . Why is  $r + (1, 1)$  not a valid vector sum here? (These are 3-vectors: same arithmetic as in  $\mathbb{R}^2$ , just one more entry.)

**W3.** Sanity check: can you form  $(4, 1) + (0, 2, 5)$ ? Can you form  $(18 \text{ cm}, 4 \text{ mm}) + (1, 1)$  as a meaningful measurement vector?

## Your turn

**Y1.** Compute

$$\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \quad \text{and} \quad (-2) \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}.$$

**Y2.** In  $\mathbb{R}^2$ , sketch  $u = (3, 1)$  and  $\frac{1}{2}u$ . In words, what did scaling by  $1/2$  do? What would  $-u$  look like?

**Y3.** A 2-vector of temperatures in  $^{\circ}\text{C}$  is  $x = (10, 14)$ . A classmate types  $x + 5$  in software and gets  $(15, 19)$ . Is that the same as a vector sum  $x + v$  for some 2-vector  $v$ ? If you need “five degrees warmer each day,” what vector should you add?

## Answers (Your turn)

Y1.  $(5, 2, 0)$  and  $(-2, -1)$ .

Y2.  $\frac{1}{2}u$  is the same direction, half as long.  $-u$  is the opposite direction, same length.

Y3. Software added the scalar 5 to each entry; that is not a vector–vector sum. Adding five degrees each day is the vector sum  $x + (5, 5)$ .